RAJALAKSHMI ENGINEERING COLLEGE

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Started on	Sunday, 30 August 2026, 8:19 PM
State	Finished
Completed on	Sunday, 30 August 2026, 8:30 PM
Time taken	11 mins 49 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two arrays array_One[] and array_Two[] of same size N. We need to first rearrange the arrays such that the sum of the product of pairs(1 element from each) is minimum. That is SUM (A[i] * B[i]) for all i is minimum.

For example:

Input	Result
3 1 2 3 4 5 6	28

Answer: (penalty regime: 0 %)

```
#include<stdio.h>

void swap(int *a,int *b){
    int temp = *a;
    *a = *b;
    *b = temp;
}

int partition(int arr[],int l,int h){
    int i = l - 1;
    int p = arr[h];

    for(int j = l; j < h; ++j){
        if(arr[j] < p){
            ++i;
            swap(&arr[i],&arr[j]);
        }
    }

    swap(&arr[h],&arr[i+1]);
    return i + 1;
}

void qs(int a[],int l,int h){
    if(l <= h){
        int p = partition(a,l,h);
        qs(a,l,p-1);
        qs(a,p+1,h);
    }
}

int main(){
    int n;
    scanf("%d",&n);

    int arr[n];
    for(int i=0;i<n;++i)
        scanf("%d",&arr[i]);

    int arr2[n];
    for(int i=0;i<n;++i)
        scanf("%d",&arr2[i]);

    qs(arr,0,n-1);
    qs(arr2,0,n-1);

    int ans = 0;

    for(int i=0;i<n;++i)
        ans += (arr[i] * arr2[n-1-i]);

    printf("%d\n",ans);
```

	Input	Expected	Got	
✔	3 1 2 3 4 5 6	28	28	✔
✔	4 7 5 1 2 1 3 4 1	22	22	✔
✔	5 20 10 30 10 40 8 9 4 3 10	590	590	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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